

Methodology

Spatiotemporal Graph Neural Networks for Real-Time Fault Localization in Smart Grids

Using the IEEE 39-Bus (New England) Test System

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Abstract

This document presents the complete research methodology for building and validating a Spatiotemporal Graph Neural Network (ST-GNN) to detect and localize electrical faults in the IEEE 39-bus (New England) test system. The methodology covers seven sequential phases: graph construction, synthetic fault-data generation, preprocessing, architecture design, model training, evaluation, and robustness analysis. The pipeline combines Graph Convolutional Networks (GCN) for spatial topology encoding with Long Short-Term Memory (LSTM) units for temporal transient modelling, achieving a targeted fault-localization accuracy exceeding 90% at the individual bus level.

1. Overview & Pipeline

The IEEE 39-bus New England system provides a realistic medium-scale transmission benchmark with 39 buses, 46 transmission lines, 10 generators, and 19 load buses. Table 1 summarises the seven-phase pipeline and Figure 1 illustrates the flow graphically.

Table 1: ST-GNN methodology pipeline.

Ph.	Activity	Description	Key Output
1	Graph Construction	Model 39-bus topology as weighted graph	$\mathcal{G}=(V, E)$, matrix $\mathbf{A}$
2	Data Generation	Simulate faults via Pandapower	$\sim 60\,000$ labeled samples
3	Preprocessing	Normalize, window, augment	$\mathbf{X} \in \mathbb{R}^{39 \times 5 \times 50}$
4	Architecture	Build GCN + LSTM model	PyG model (~ 1.4 M params)
5	Training	Optimize with Adam + multi-task loss	Trained weights
6	Evaluation	Accuracy, F1-score, ablation	Benchmark table
7	Robustness	Noise, N-1, transfer learning	Degradation curves

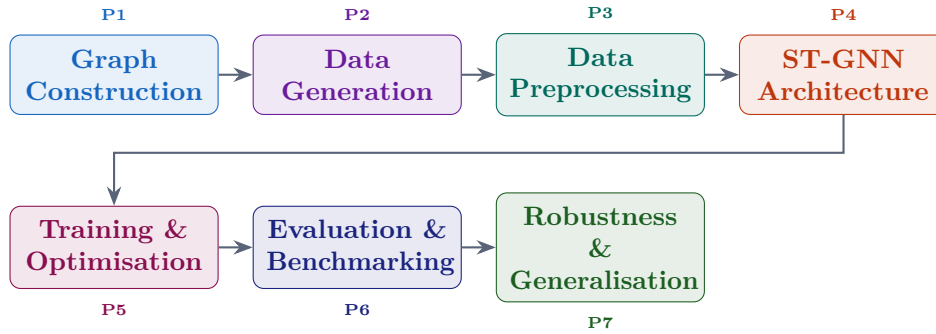


Figure 1: Seven-phase methodology pipeline for ST-GNN fault localization in the IEEE 39-bus system. Phases 1–4 build the model; Phases 5–7 train, evaluate, and stress-test it.

2. Phase 1 — Power System Model and Environment

2.1 Base Topology and Grid Representation

The grid topology serving as the benchmark for this study is the IEEE 39-bus (New England) transmission system, a widely recognized standard for medium-scale power system analysis. To generate realistic steady-state operational data and dynamic fault transients, the Python-based `pandapower` library is employed as the primary simulation engine.

Within this computational environment, the grid is mathematically represented as an interconnected network, or weighted undirected graph $\mathcal{G} = (V, E)$, comprising 39 nodes (buses) linked by high-voltage transmission lines. The `pandapower` framework facilitates the accurate resolution of Newton-Raphson power flows and standard short-circuit computations, enabling the localized extraction of bus features that inherently respect the physical laws and topological limits of the grid.

3. Phase 2 — Synthetic Data Generation & Simulation

3.1 Fault Scenario Generation

To ensure the proposed GNN is capable of comprehensive multi-class anomaly detection, four distinct electrical fault types are simulated (detailed in Table 2): Single Line-to-Ground (SLG), Line-to-Line (LL), Double Line-to-Ground (DLG), and symmetric Three-Phase (3PH) faults.

Table 2: Simulated fault types and their respective properties.

Type	Abbrev.	Description	Prevalence
Single Line-to-Ground	SLG	One phase contacts ground (most common)	$\approx 70\%$
Line-to-Line	LL	Two phases contact each other	$\approx 15\%$
Double Line-to-Ground	DLG	Two phases contact ground simultaneously	$\approx 10\%$
Three-Phase Symmetric	3PH	All three phases short-circuit (most severe)	$\approx 5\%$

A rigorous dynamic parameter sweep is executed to guarantee dataset diversity and structural robustness. Specifically, faults are systematically simulated at every individual bus within the network (buses 1 through 39) across a highly varied set of operational conditions. These conditions encompass sweeping fault resistances across 0, 5, 10, and 20 Ω , alongside scaling the global active and reactive load levels to 0.6, 1.0, and 1.3 per-unit (p.u.).

Implementing these exhaustive parameter sweeps is critical for model generalization; it prevents the neural network from overfitting to ideal zero-impedance faults under nominal loads, forcing the algorithm to learn underlying topological and physical fault signatures that remain resilient under varying grid stresses and high-impedance anomaly conditions. For every unique fault simulation generated through this sweep, network state parameters are extracted over a sliding temporal window consisting of exactly 50 timesteps.

3.2 Normal Operation and Noise Injection

In addition to transient anomalous states, establishing a comprehensive baseline of healthy grid behavior is imperative for binary fault detection (differentiating between standard operations and active faults). To fulfill this requirement, 10 000 independent normal operation samples are generated.

To accurately simulate the stochastic nature of physical power grids and the inherent measurement uncertainties associated with real-world Phasor Measurement Units (PMUs), random Gaussian noise ($\mu = 0, \sigma = 0.005$) is systematically injected into the standard operational features. This noise injection persists across the 50-timestep windows, serving as an essential data augmentation strategy that fortifies the model against false positive classifications triggered by routine sensor variance.

4. Phase 3 — Feature Extraction & Dataset Structure

4.1 Temporal Feature Extraction

For each simulation timestep and across every bus, a 5-dimensional temporal feature vector is extracted to construct the localized node attributes. The captured grid parameters are summarized in Table 3.

Table 3: Node feature set (five features extracted per bus, per timestep).

#	Feature	Symbol	Physical Meaning
1	Voltage Magnitude	V_m	Deviation from nominal 1.0 p.u.
2	Voltage Phase Angle	θ	Phase shift at the bus (degrees)
3	Active Power	P	Net generation minus load (MW)
4	Reactive Power	Q	Reactive power balance (MVAR)
5	Frequency Deviation	Δf	Deviation from 60 Hz nominal (Hz)

To mimic realistic frequency fluctuations and mechanical transient perturbations, the Frequency Deviation (Δf) is uniquely simulated with a distinct, independent synthetic noise profile ($\mu = 0, \sigma = 0.01$).

4.2 Final Data Matrix and Label Definition

Following the completion of the simulation pipeline, the resulting synthetic dataset is aggregated into a standardized multi-dimensional tensor. The final data matrix is structured with the shape $(N_{\text{samples}}, 39, 5, 50)$, corresponding respectively to the total number of samples, 39 buses, 5 extracted features, and 50 discrete timesteps.

Finally, to accommodate downstream multi-task learning, two explicit labels are mapped to each temporal window sample:

- **Localization Label (y_{loc}):** An integer value ranging from 0 to 38 representing the specific index of the faulted bus, or -1 designating standard normal operation.
- **Classification Label (y_{type}):** An integer index corresponding to the discrete state of the system, where 0 = SLG, 1 = LL, 2 = DLG, 3 = 3PH, and 4 = Normal.

5. Phase 4 — ST-GNN Architecture

5.1 Architecture Overview

The proposed **Spatiotemporal Graph Convolutional Network (ST-GCN)** processes $\mathbf{X}_t \in \mathbb{R}^{39 \times 5 \times 50}$ through two stacked GCN layers (spatial), a two-layer LSTM (temporal), global mean pooling, and two parallel output MLP heads.

5.2 Spatial Block: Graph Convolutional Network

For each timestep, the layer-wise propagation rule is

$$\mathbf{H}^{(\ell+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(\ell)} \mathbf{W}^{(\ell)}\right), \quad (1)$$

where $\mathbf{H}^{(\ell)} \in \mathbb{R}^{39 \times d_\ell}$ are node embeddings at layer ℓ , $\mathbf{W}^{(\ell)}$ is the trainable weight matrix, and $\sigma(\cdot) = \text{ReLU}(\cdot)$. Two stacked GCN layers provide a 2-hop receptive field.

Table 4: GCN spatial block dimensions.

Layer	Input dim	Output dim	Receptive field
GCN-1	5	64	1-hop neighbours
GCN-2	64	128	2-hop neighbours

5.3 Temporal Block: LSTM

The per-timestep GCN embeddings $\mathbf{H}_1, \dots, \mathbf{H}_{50} \in \mathbb{R}^{39 \times 128}$ are processed node-independently by a 2-layer LSTM with hidden size 256 and inter-layer dropout $p = 0.3$. The final hidden state $\mathbf{h}_T \in \mathbb{R}^{39 \times 256}$ encodes the full spatiotemporal fault signature.

5.4 Readout & Output Heads

Global mean pooling yields $\mathbf{g} \in \mathbb{R}^{256}$. Two parallel heads operate on $\mathbf{g}$:

- **Localization head:** MLP $256 \rightarrow 128 \rightarrow 64 \rightarrow 39$ with Softmax; arg max gives the predicted faulted bus.
- **Fault-type head:** MLP $256 \rightarrow 64 \rightarrow 4$ with Softmax; classifies $\{\text{SLG, LL, DLG, 3PH}\}$.

Table 5: Complete ST-GNN architecture summary.

Component	Type	Output shape	Params
Input	Node feature tensor	(39, 5, 50)	—
GCN-1 (per timestep)	Graph conv. + ReLU	(39, 64, 50)	320
GCN-2 (per timestep)	Graph conv. + ReLU	(39, 128, 50)	8 192
LSTM	2-layer, hidden=256	(39, 256)	1 311 744
Global mean pool	Aggregation	(256,)	0
Localization MLP	FC + Softmax	(39,)	49 191
Type MLP	FC + Softmax	(4,)	17 412
Total			≈ 1.39 M

6. Phase 5 — Model Training & Optimization

6.1 Multi-Task Loss Function

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{loc}} + (1 - \alpha) \mathcal{L}_{\text{type}}, \quad \alpha = 0.7, \quad (2)$$

where both terms are weighted cross-entropy losses. The weighting $\alpha = 0.7$ prioritizes localization precision, which is operationally more critical for maintenance dispatching.

6.2 Training Configuration

Table 6: Training hyperparameters.

Hyperparameter	Value	Rationale
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)	Adaptive lr
Initial learning rate	1×10^{-3}	Standard for GNNs
LR scheduler	ReduceLROnPlateau ($\times 0.5$, patience= 10)	Prevents stagnation
Batch size	32	GPU memory balance
Max epochs	200	With early stopping
Early-stop patience	20	Validation F1
Dropout (LSTM / MLP)	0.3 / 0.2	Regularization
Weight decay (L_2)	1×10^{-4}	Overfitting control
Random seed	42	Reproducibility

6.3 Baseline Models

1. **Threshold rule system** — SCADA-style voltage threshold ($|V_m - 1.0| > 0.1$ p.u.).
2. **Random Forest** — trained on flattened node features, topology-agnostic.
3. **CNN-LSTM** — same temporal depth, no graph structure.
4. **Vanilla GCN** — graph convolution only, no temporal modelling.

7. Phase 6 — Evaluation & Benchmarking

7.1 Evaluation Metrics

Table 7: Primary evaluation metrics and targets.

Metric	Definition	Target
Fault Detection Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	$> 95\%$
Top-1 Localization Acc.	Correct bus / total samples	$> 90\%$
Top-3 Localization Acc.	Correct bus in top-3 predictions	$> 98\%$
Weighted F1 (fault type)	Harmonic mean of precision & recall	> 0.92
Mean Abs. Bus Error	$\mathbb{E}[\hat{y}_{loc} - y_{loc}]$	< 2
Inference Latency	Wall-clock time per sample (CPU)	< 10 ms

7.2 Ablation Study

Table 8: Ablation experiments and expected performance impact.

Exp.	Removed	Motivation	Expected Δ
A1	LSTM (spatial only)	Isolate temporal value	$\downarrow$ 8–12%
A2	GCN (temporal only)	Isolate graph-structure value	$\downarrow$ 15–20%
A3	Impedance weighting	Binary vs. weighted $\mathbf{A}$	$\downarrow$ 3–5%
A4	Fault-type head (single task)	Multi-task benefit	$\downarrow$ 2–4%
A5	Noise augmentation	Robustness benefit	Noisy-set drop

A 39×39 confusion matrix is generated to verify that localization errors occur at electrically adjacent buses, confirming physically plausible behaviour.

8. Phase 7 — Robustness & Generalization Analysis

Noise sensitivity. The model is evaluated on test sets with additional Gaussian noise $\sigma \in \{0.0, 0.01, 0.02, 0.05, 0.10\}$ p.u., producing a noise–accuracy degradation curve.

High-impedance faults. Detection performance is specifically measured for $R_f > 20 \Omega$, the most challenging regime for fault identification.

N-1 contingency testing. One transmission line is removed from the network without retraining, assessing topology-change generalization.

Cross-system transfer (IEEE 14-bus). The 39-bus model is fine-tuned on a small 14-bus labeled dataset to assess transfer of learned physical representations.

9. Implementation Details

Table 9: Software environment.

Tool / Library	Version	Purpose
Python	3.9	Primary programming language
PyTorch	2.1.0	Deep learning framework
PyTorch Geometric	2.4.0	GNN layers (GCNConv)
Pandapower	2.13.1	Power-flow simulation & fault generation
NumPy / SciPy	1.24 / 1.11	Numerical computation
Scikit-learn	1.3	Baseline models & evaluation metrics
Weights & Biases	0.16	Experiment tracking
Matplotlib	3.7	Result visualization

Reproducibility is ensured by fixing all random seeds (`torch.manual_seed(42)`, `numpy.random.seed(42)`), logging all runs with Weights & Biases, and providing a Docker container with pinned dependency versions.

10. Project Timeline

Table 10: 14-week project schedule.

Week(s)	Activity	Deliverable
1–2	Grid modeling + Pandapower setup	IEEE 39-bus graph + adjacency matrix
3–4	Fault simulation + dataset generation	~60 K labeled CSV samples
5	Preprocessing + data loader	PyG Data objects, splits
6–7	ST-GNN implementation	PyTorch model code
8–9	Training + hyperparameter tuning	Trained model checkpoint
10	Baseline experiments	Comparison results table
11	Ablation studies	Component contribution analysis
12	Robustness testing	Noise–accuracy curves
13	Visualization + confusion matrix	Figures and heatmaps
14	Report writing + code cleanup	Final research paper

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